

Table 7.5: Pre/post changes in PDC-related experience. Percent favorable is the percentage of responses coded 5 or higher.

Construct	Pre M	Post M	ΔM	Pre %	Post %	Δ pp	d
Parallel processing	1.50	2.25	0.75	0.0	25.0	25.0	0.54
Distributed computing	1.75	2.12	0.38	0.0	25.0	25.0	0.26
Event handling	1.62	2.25	0.62	0.0	25.0	25.0	0.43

Survey and Other Anonymized Data and Analysis

Survey responses were analyzed using a seven-point ordinal scale. For experience items, responses were coded from 1 = no experience to 7 = extremely experienced. For agreement items, responses were coded from 1 = strongly disagree to 7 = strongly agree. Percent favorable was calculated as the percentage of students selecting 5 or higher on the seven-point scale. Because the pre- and post-surveys were anonymous and analyzed as aggregate responses, the results are reported as group-level pre/post changes rather than matched individual gains.

Overall, the data show that students entered the course with relatively strong general programming confidence but limited exposure to parallel and distributed computing (PDC). The composite mean across the three PDC-related experience items increased from 1.62 on the pre-survey to 2.21 on the post-survey, a gain of 0.58 points on the seven-point scale. The percentage of favorable responses increased from 0% to 25% for parallel processing, distributed computing, and event handling. These results suggest that the course introduced students to PDC concepts that were largely unfamiliar at the beginning of the semester.

The strongest quantitative gains occurred in core COSC 1560 programming topics that support later PDC learning. The composite programming-experience mean across ten C++ topics increased from 4.17 to 5.26, for an average gain of 1.08 points. The largest individual gains were observed for C++ structures ($\Delta M = 2.73$), classes and objects ($\Delta M = 1.93$), arithmetic expressions ($\Delta M = 1.09$), pointers ($\Delta M = 0.98$), branching ($\Delta M = 0.84$), and arrays ($\Delta M = 0.80$). These results are important because arrays, pointers, structures, and object-oriented design provide the technical foundation for later examples involving data decomposition, memory organization, and task-based problem solving.

Student attitudes toward computing remained generally positive. In the post-survey, 87.5% of students rated their interest in computing as favorable, and 87.5% rated their general understanding of computing as favorable. The PDC-specific attitudinal measures were more conservative: 50.0% gave favorable ratings for interest in PDC, 50.0% gave favorable ratings for the importance of PDC, and 37.5% gave favorable ratings for understanding of PDC. This pattern is consistent with a course design in which PDC is introduced as an embedded CS1/CS2-level topic rather than as a full dedicated parallel computing course; students gained initial exposure, but their self-reported mastery remained lower than their general programming confidence.

Perceived helpfulness of course activities. Post-survey responses also indicate that students viewed the main instructional activities as helpful. Programming assignments re-

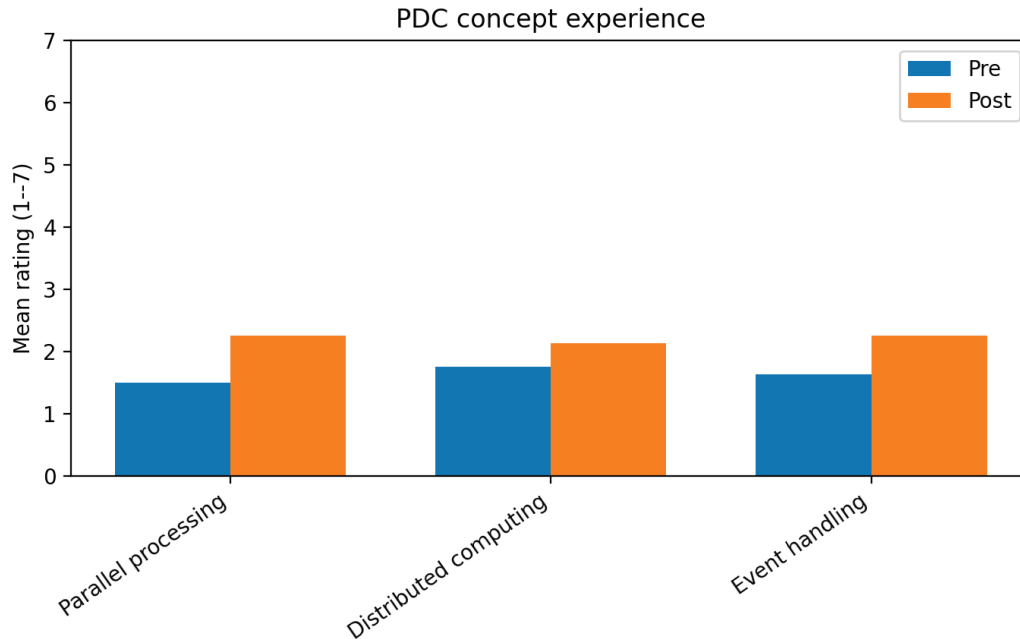


Figure 7.3: Pre/post comparison of student self-reported experience with PDC-related concepts. Mean ratings are reported on a seven-point scale.

ceived the highest rating ($M = 6.62$), with 100% of responses favorable and 100% rated very helpful or extremely helpful. Attending class was similarly strong ($M = 6.50$), followed by in-class practices ($M = 6.12$), textbook use ($M = 5.62$), exam preparation ($M = 5.62$), and meeting with the instructor outside of class ($M = 5.62$). These results support the use of a layered instructional sequence: direct instruction, in-class practice, programming assignments, and assessment preparation.

Post-course learning outcomes. Students reported high confidence on the major COSC 1560 learning outcomes. All students gave favorable ratings for writing C++ programs using arrays, pointers, structured data, and file operations; designing programs using department style guidelines; understanding the array–pointer relationship; reading and writing text/binary files; and using constructors and destructors. The strongest post-course outcome means were for arrays/pointers/structures/files ($M = 6.12$), department style guidelines ($M = 6.00$), and array–pointer relationships ($M = 6.00$). These outcomes align well with the PDC module because PDC examples depend on students’ ability to reason about data layout, repeated operations over collections, and decomposition of work into smaller units.

The survey results indicate that students showed measurable gains in PDC-related familiarity and larger gains in the programming foundations needed to understand PDC examples. The post-survey activity ratings further suggest that students benefited most from active and practice-oriented components, especially programming assignments, class attendance, and in-class practice.

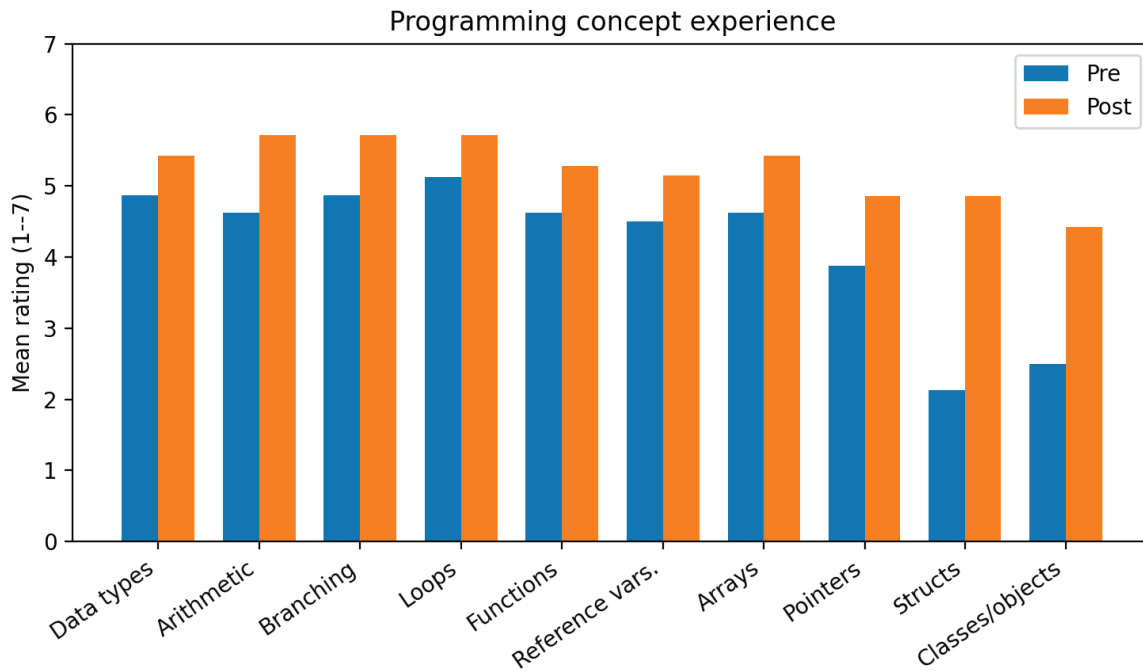


Figure 7.4: Pre/post comparison of student self-reported programming experience across core C++ topics.

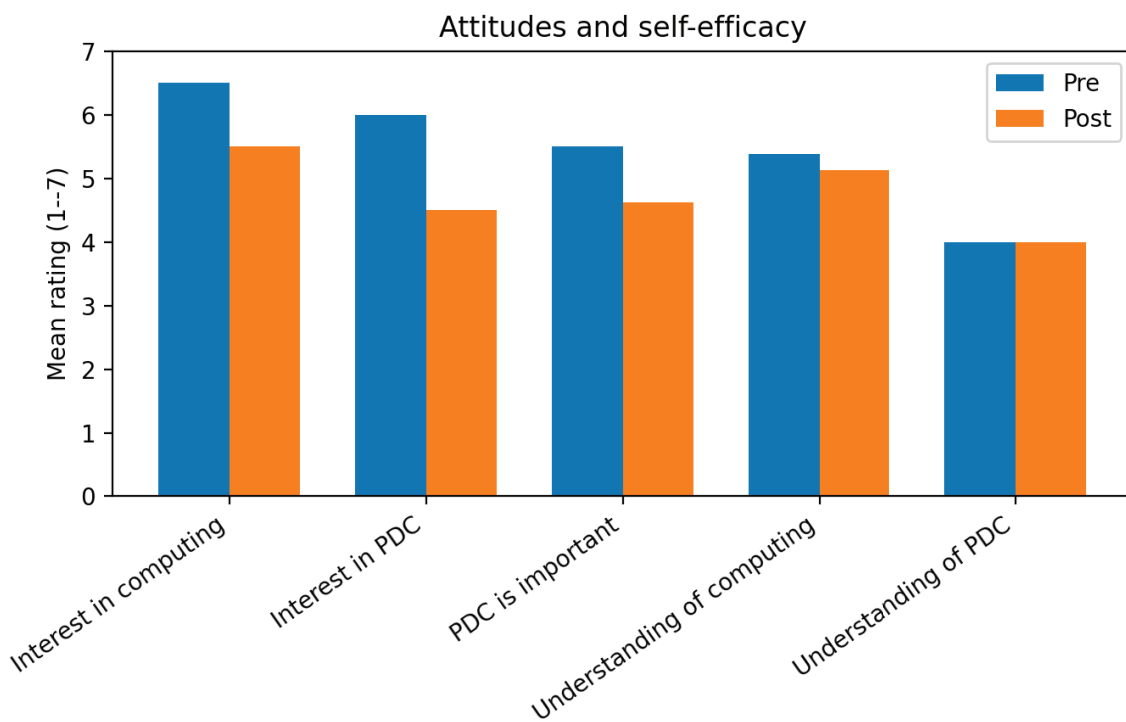


Figure 7.5: Pre/post comparison of student attitudes, interest, and self-reported understanding.

Table 7.6: Pre/post changes in programming experience across core C++ concepts.

Construct	Pre M	Post M	ΔM	Pre %	Post %	Δ pp	d
Data types	4.88	5.43	0.55	62.5	85.7	23.2	0.52
Arithmetic	4.62	5.71	1.09	62.5	85.7	23.2	0.81
Branching	4.88	5.71	0.84	75.0	85.7	10.7	0.80
Loops	5.12	5.71	0.59	87.5	100.0	12.5	0.66
Functions	4.62	5.29	0.66	62.5	85.7	23.2	0.78
Reference vars.	4.50	5.14	0.64	62.5	85.7	23.2	0.89
Arrays	4.62	5.43	0.80	75.0	71.4	-3.6	0.85
Pointers	3.88	4.86	0.98	37.5	57.1	19.6	0.76
Structs	2.12	4.86	2.73	12.5	57.1	44.6	2.22
Classes/objects	2.50	4.43	1.93	25.0	42.9	17.9	1.28

Table 7.7: Pre/post changes in interest, perceived importance, and self-reported understanding.

Construct	Pre M	Post M	ΔM	Pre %	Post %	Δ pp	d
Interest in computing	6.50	5.50	-1.00	100.0	87.5	-12.5	-0.88
Interest in PDC	6.00	4.50	-1.50	100.0	50.0	-50.0	-1.20
PDC is important	5.50	4.62	-0.88	75.0	50.0	-25.0	-0.64
Understanding of computing	5.38	5.12	-0.25	87.5	87.5	0.0	-0.26
Understanding of PDC	4.00	4.00	0.00	37.5	37.5	0.0	0.00

Table 7.8: Post-survey ratings of course learning activities.

Activity	Mean	Favorable %	Very/Extremely Helpful %
Textbook	5.62	100.0	50.0
Programming assignments	6.62	100.0	100.0
In-class practices	6.12	100.0	87.5
Attending classes	6.50	100.0	100.0
Meeting with the instructor outside of class	5.62	62.5	62.5
Tutoring provided by REEG ARC peer tutors	4.38	25.0	12.5
Exams and preparing for exams	5.62	100.0	50.0

Table 7.9: Post-survey ratings of course learning outcomes.

Learning outcome	Mean	Favorable %	Agree/Strongly Agree %
Arrays/pointers/structs/files	6.12	100.0	87.5
Department style guidelines	6.00	100.0	62.5
Array–pointer relationship	6.00	100.0	75.0
Character/string functions	5.38	87.5	50.0
Text/binary files	5.88	100.0	75.0
ADTs and structures	5.25	62.5	50.0
Classes with public/private members	5.62	87.5	62.5
Constructors/destructors	5.62	100.0	50.0